

Colony Streaking

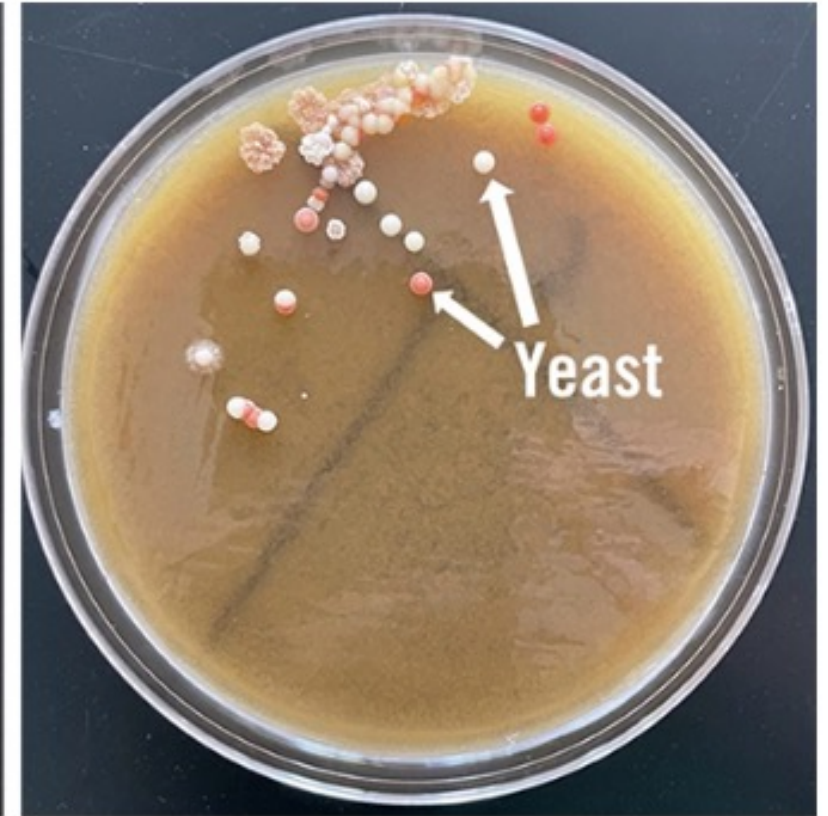
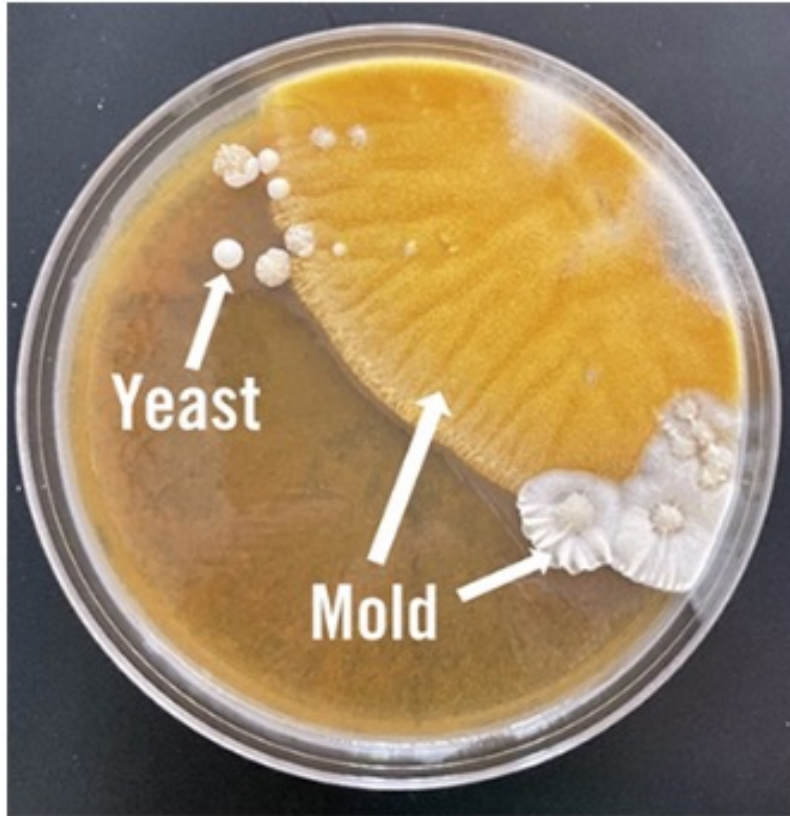
LaBella Lab
Wild Yeast Project

Outline

- Colony morphology 101
- Experimental design – antifungal experiment
- Protocol review
- Preparing notebooks



What's on a plate?



Bacterial Colonies

- Smaller than yeast colonies (*typically*)
- Yellow or cream colored
- Shine on top of colony (*typically*)

***Do we expect bacteria
on our plates?***



Bacillus subtilis(3)



Proteus vulgaris(4)



Staphylococcus aureus(5)

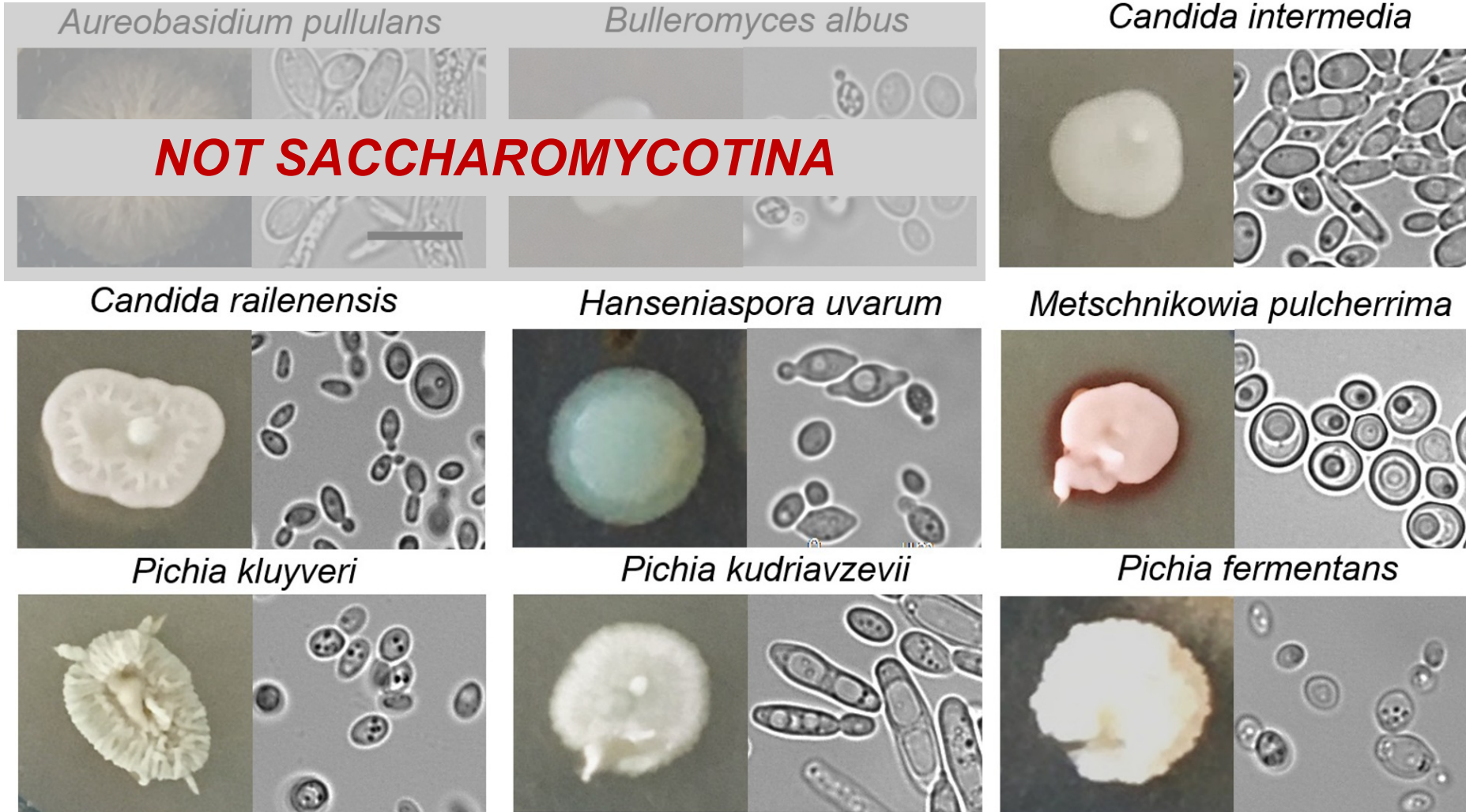


Yeast Colonies

- Matte round colonies (*typically*)
- Larger than bacterial colonies

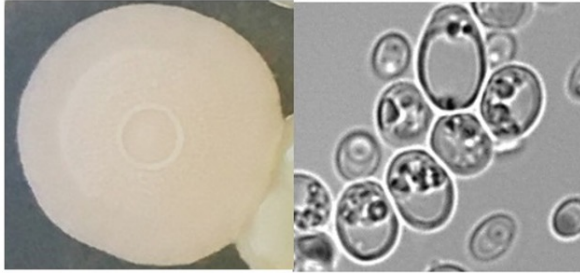


But wait, there's more... morphologies, that is!

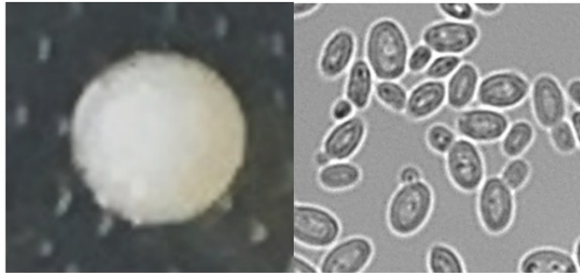


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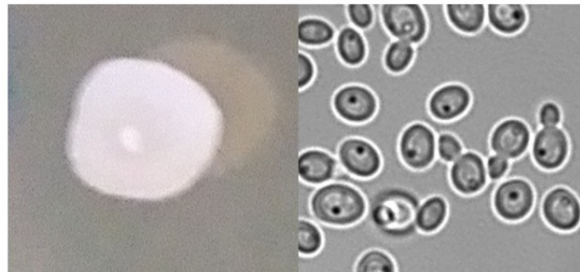
Pichia terricola



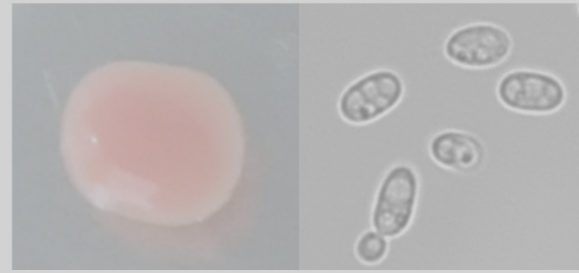
Starmerella bacillaris



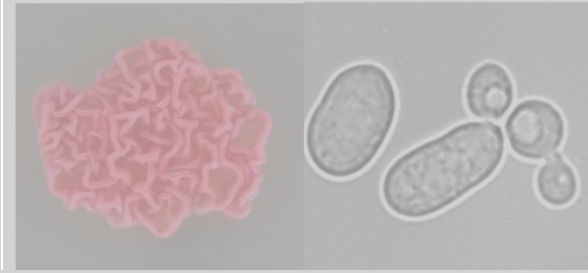
Wickerhamomyces anomalus



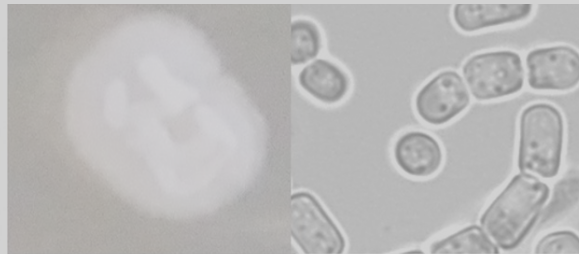
Rhodotorula glutinis



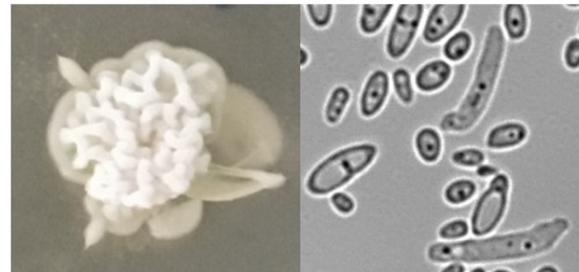
Sporobolomyces roseus



NOT SACCHAROMYCOTINA



Zygoascus meyeriae



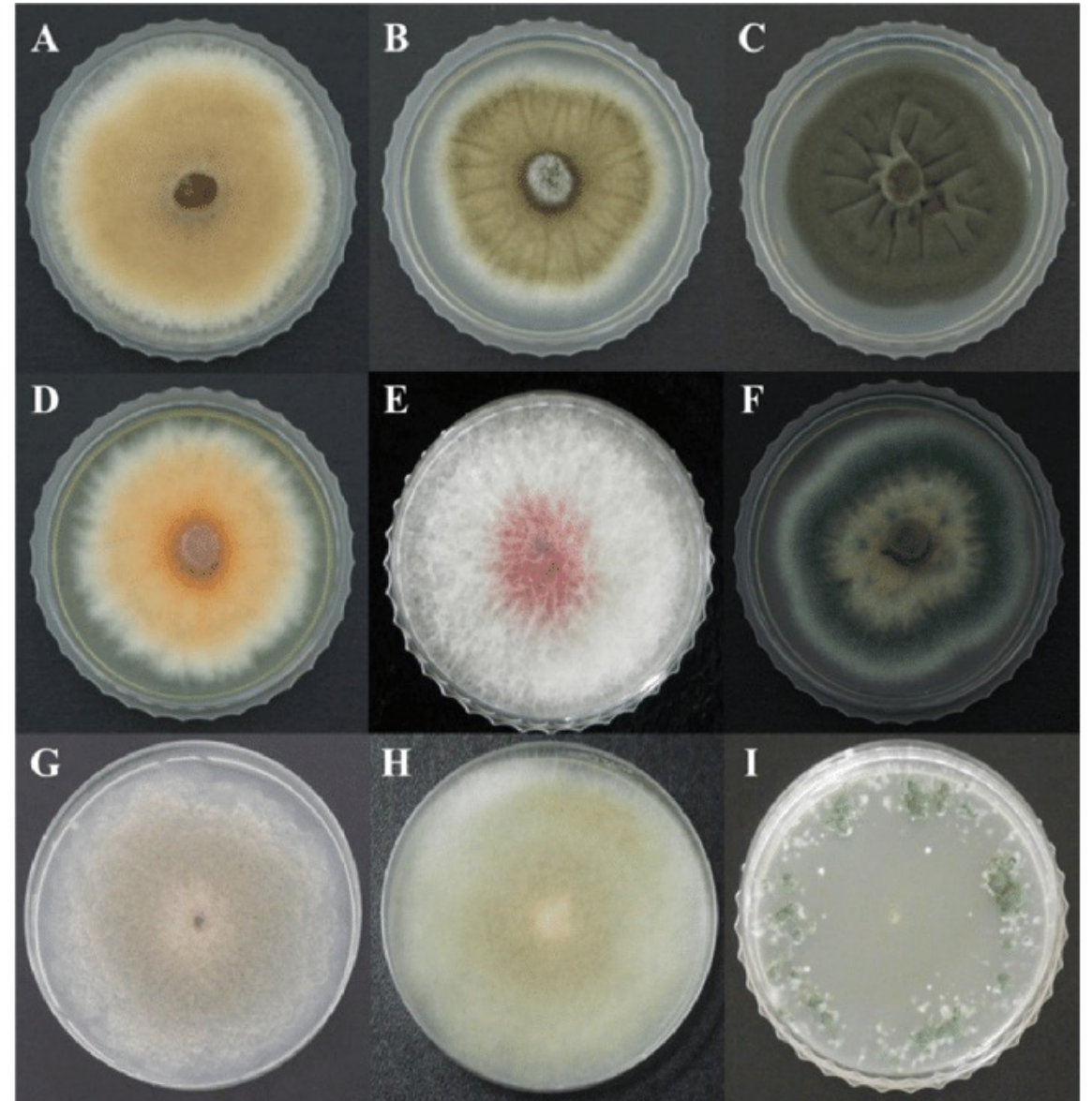
Red Yeasts

We don't want to sequence
Rhodotorula sp. (on purpose)



Filamentous fungal fellows

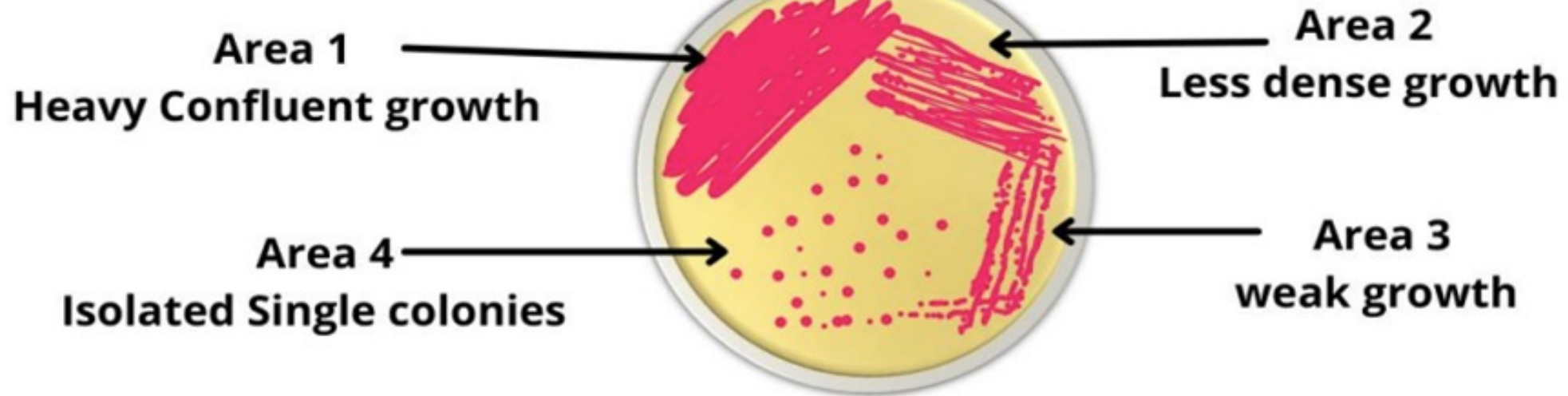
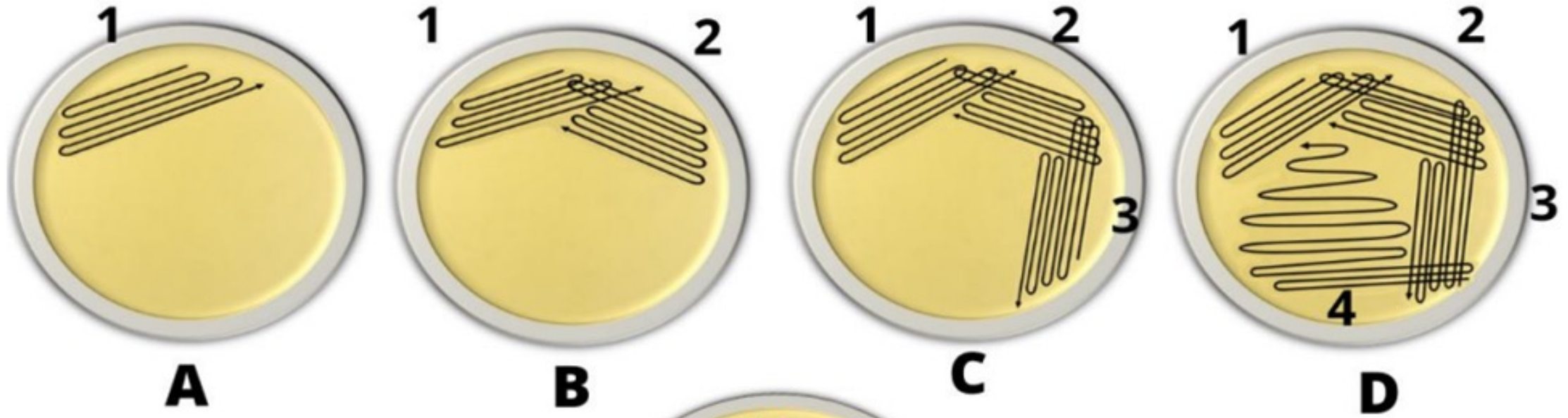
Stay as far away from these as possible when selecting/picking yeast colonies



Yeast Spotting Practice

<https://www.rowleylab.com/identifying-yeast-agar>





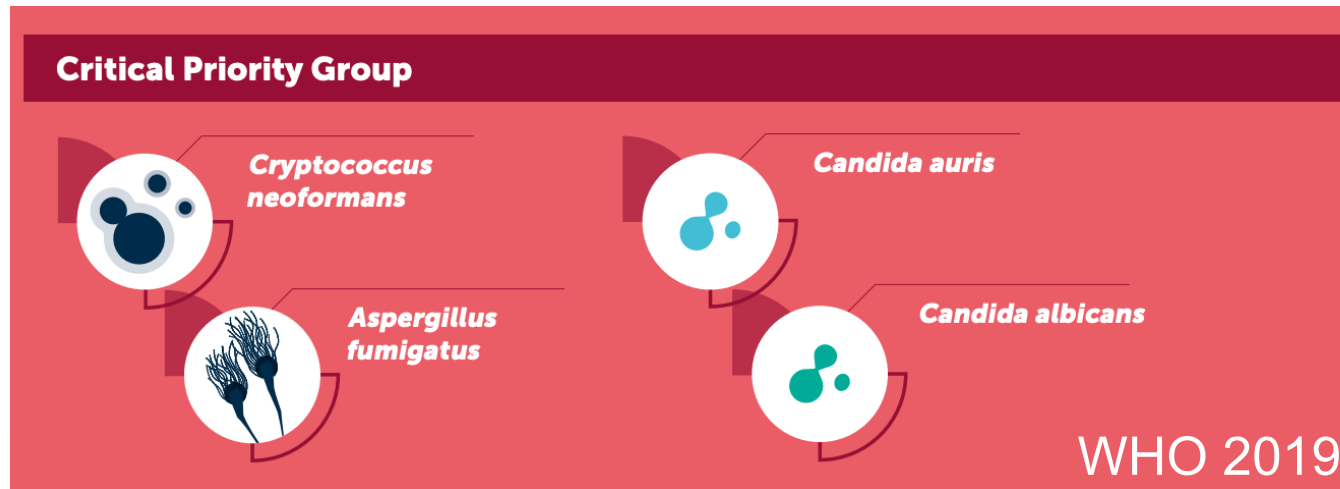
Antifungal Resistance

Antifungal resistant strains of fungal pathogens are becoming more prevalent
Both crop and human pathogens

Global emergence of *Candidozyma auris*



Adapted from Rhodes & Fisher (2019) Curr Opin Microbio



NEXT WEEK
Experimental Design



Antifungal Resistance

Antifungal resistant strains of fungal
pathogens

Both *Candida albicans*

Idozyma auris



Curr Opin Microbio

Are the wild yeast strains we isolated
antifungal resistant?

Critical Prior



*Aspergillus
fumigatus*



Candida albicans



NEXT WEEK
Experimental Design



Types of Antifungals

Azoles

Most commonly used

Targets ergosterol biosynthesis pathway (cell membrane)



Echinocandins

Targets fungal-specific beta-glucan biosynthesis (cell wall)

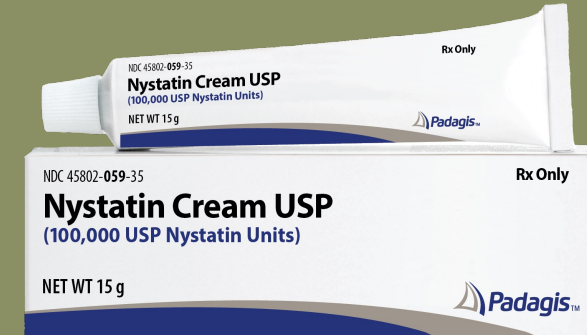
Most effective against *Aspergillus* sp., *Candida* sp.



Polyenes

Bind and remove ergosterol from membranes, changing cell permeability

Higher human toxicity; used for more serious infections



NEXT WEEK
Experimental Design



Experimental Design

- What is being tested? What is an appropriate control?
- Are certain antifungals better than others?
- Are there other products that might inhibit fungal growth?



Colony Streaking Protocol

- Step 1: Observations
 - What do you observe about each sample?
 - How does it look?
 - What's the overall content? How do colonies look? Does it look like there are other organisms present?
 - What does the sample smell like?
 - *Never smell a sample directly*; instead, hold the plate away from your body and use a hand to gently fan or waft the smell towards you.
- Step 2: Identifying yeast colonies
 - Circle up to 2 yeast colonies per plate
- Step 3: Plating
 - Label plates and streak individual colonies onto new YPD plate



Notebook Prep

Step 1 - Sample Observation

For each of your samples, observe the following details and report the results. *Be sure to leave room in your notebook between the questions*

1. **Overall Content:** Describe the overall content of your plate. Include descriptions of color, texture, and density.
 2. **Yeast Colonies:** Are there any yeast colonies? These colonies will be small, white-to-cream colored and typically have smooth, defined edges. Describe the appearance of these colonies
 3. **Bacteria, red yeasts, or filamentous fungi:** Are there any other types of organisms present? Describe how you came to that conclusion?
 4. **Smell:** Do NOT smell your sample directly. Hold the plate slightly away from your body, then use your hand to gently wave or fan the smell toward your nose. What does your sample smell like?
- Write out observation prompts (see handout; leave room for each of your samples)
 - Write down each step of the protocol (see handout)



Lab Work – Groups of 3

Confirm w/ Dr. Ward or Dr. LaBella that you
(and your notebook) are ready to go.

